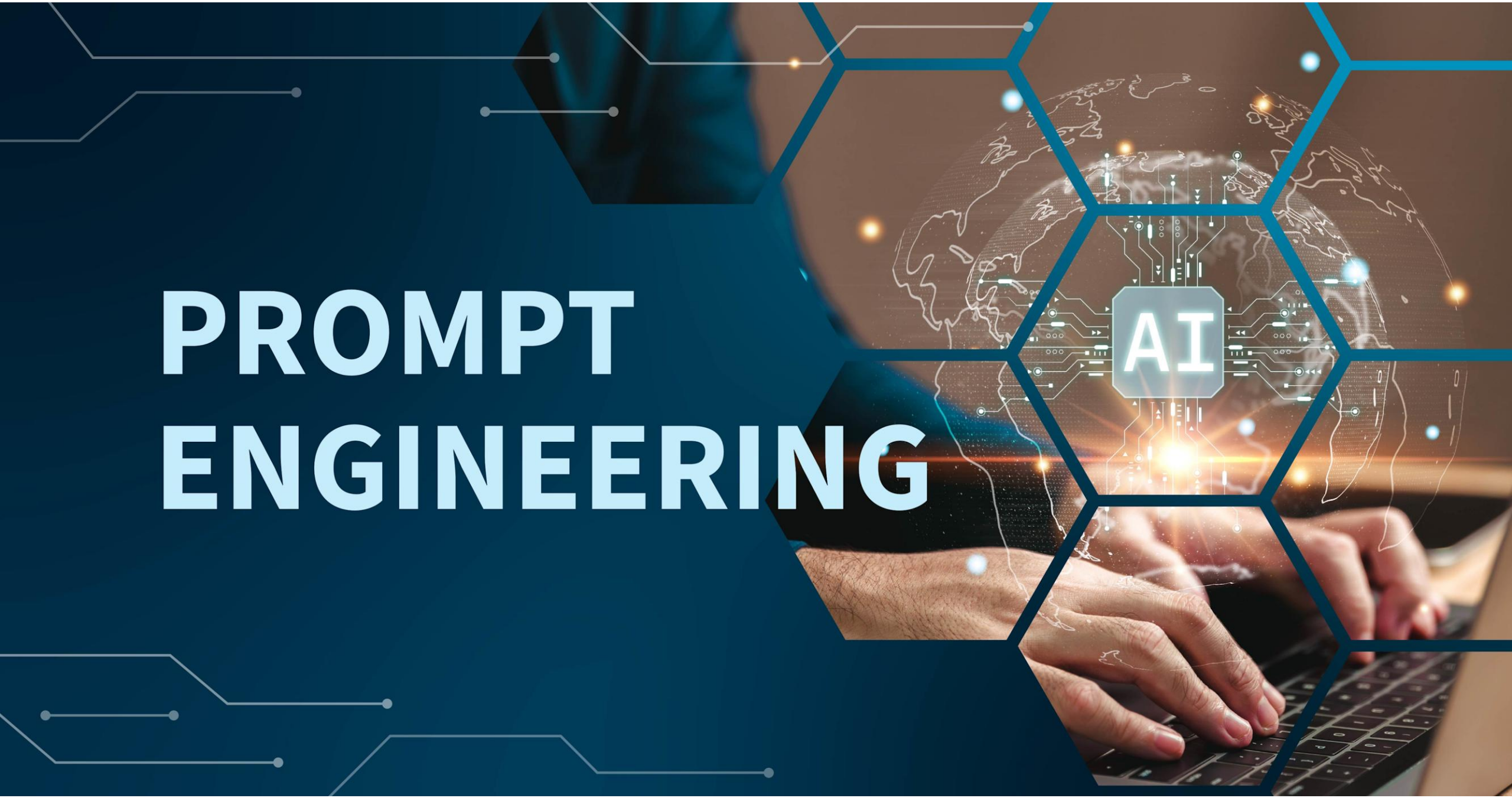
PROMPT ENGINEERING

Prompt Engineering & Building GenAI Applications

Designing and optimizing effective prompts — from anatomy and reasoning techniques to structured outputs, chaining, and evaluation tooling.

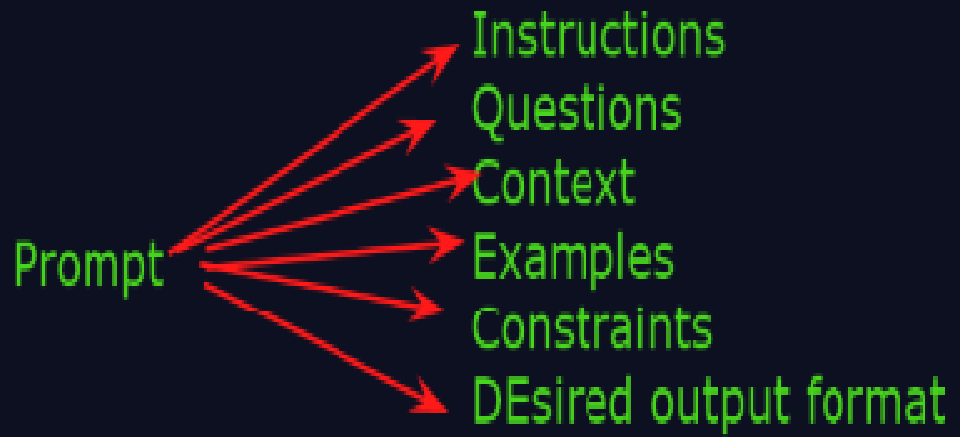
Zero/Few-Shot | Chain-of-Thought | Role Prompting | Prompt Chaining | JSON/XML Output

PROMPT ENGINEERING



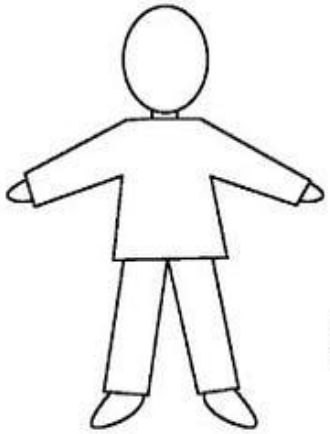
Prompt Engineering:

-It is a process of designing, structuring, and refining instruction (prompts) given to a LLM to obtain accurate, relavant and high-quality responses.



Improves the quality
reduces hallocinations
saves time
controls responses with formatting
Enables automation

You are screen sharing
Stop share



Human

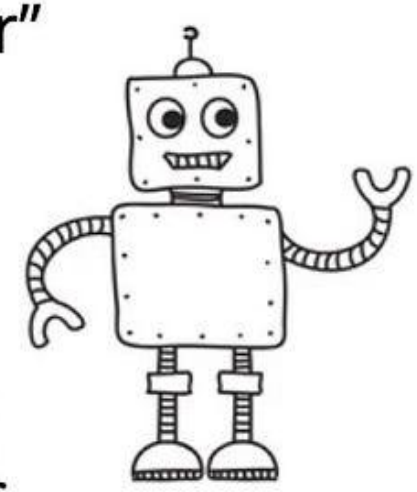
Specification of goals

"Program"

"Prompt Engineering"

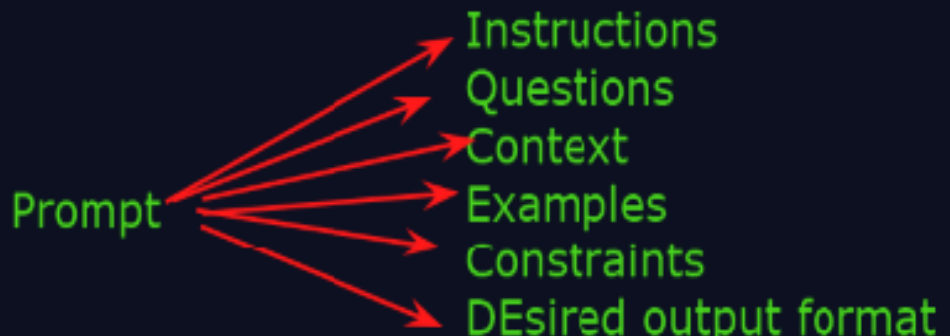
Interpretation of goals

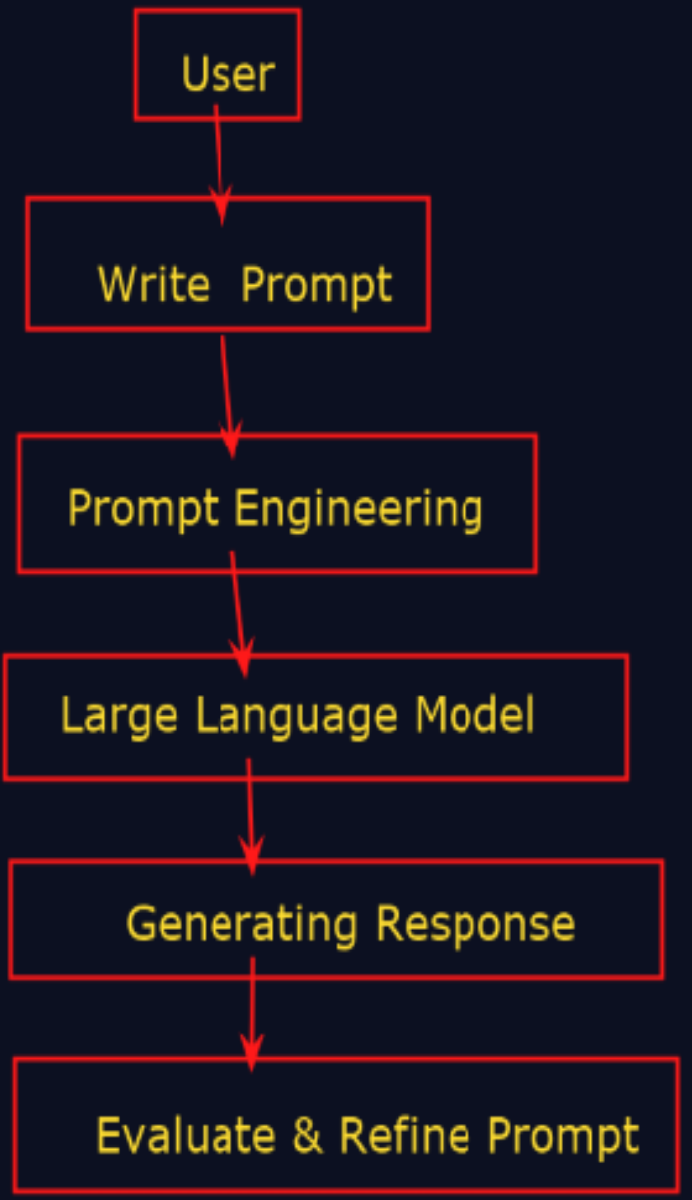
"Compiler"



LLM Taking Actions

to obtain accurate, relavant and high-quality responses.





You are screen sharing
Stop share

What is Prompt Engineering?



Prompt Engineering

involves



Crafting and Refining Prompts

to



Generate Accurate Responses

Key Principles:



Be Clear and Specific



Give Context



Set Format and Style

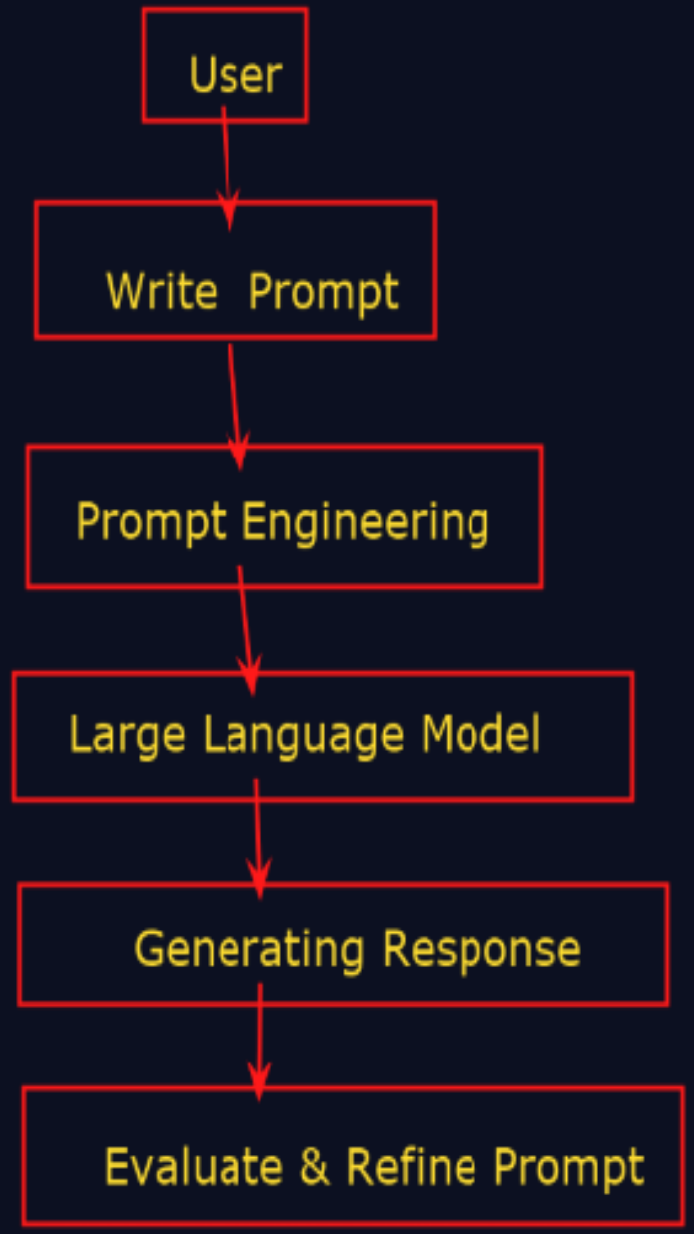


Show Examples

to obtain accurate, relavant and high-quality responses.

Good Prompt contains:

Role
Task
Context
Constraints
Input data
Output format
Examples



Key Concepts of Prompt Engineering

Token

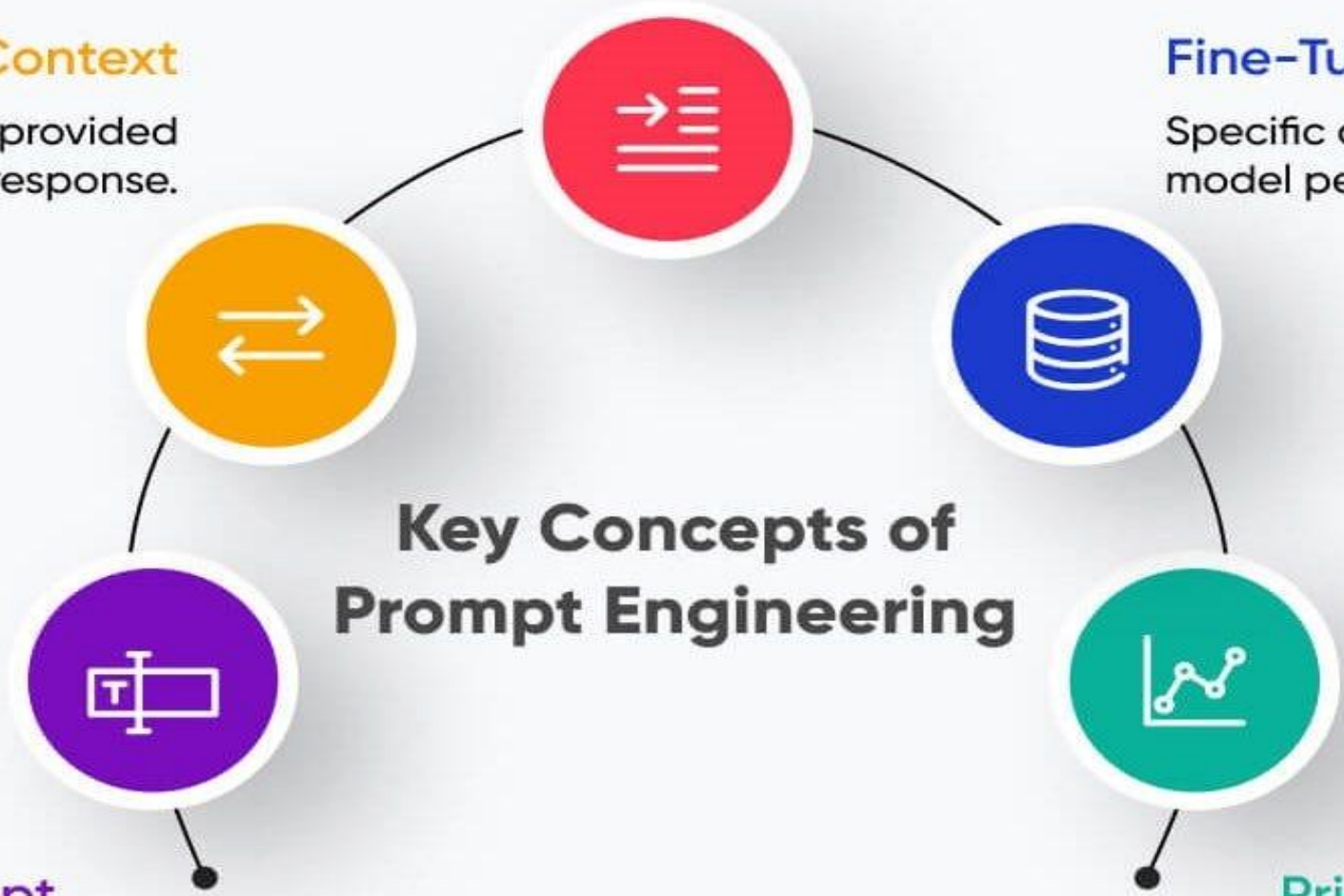
NLP has a unit of text to process inputs and generate responses.

Context

Relevant background provided to influence the response.

Fine-Tuning

Specific datasets to improve AI model performance.



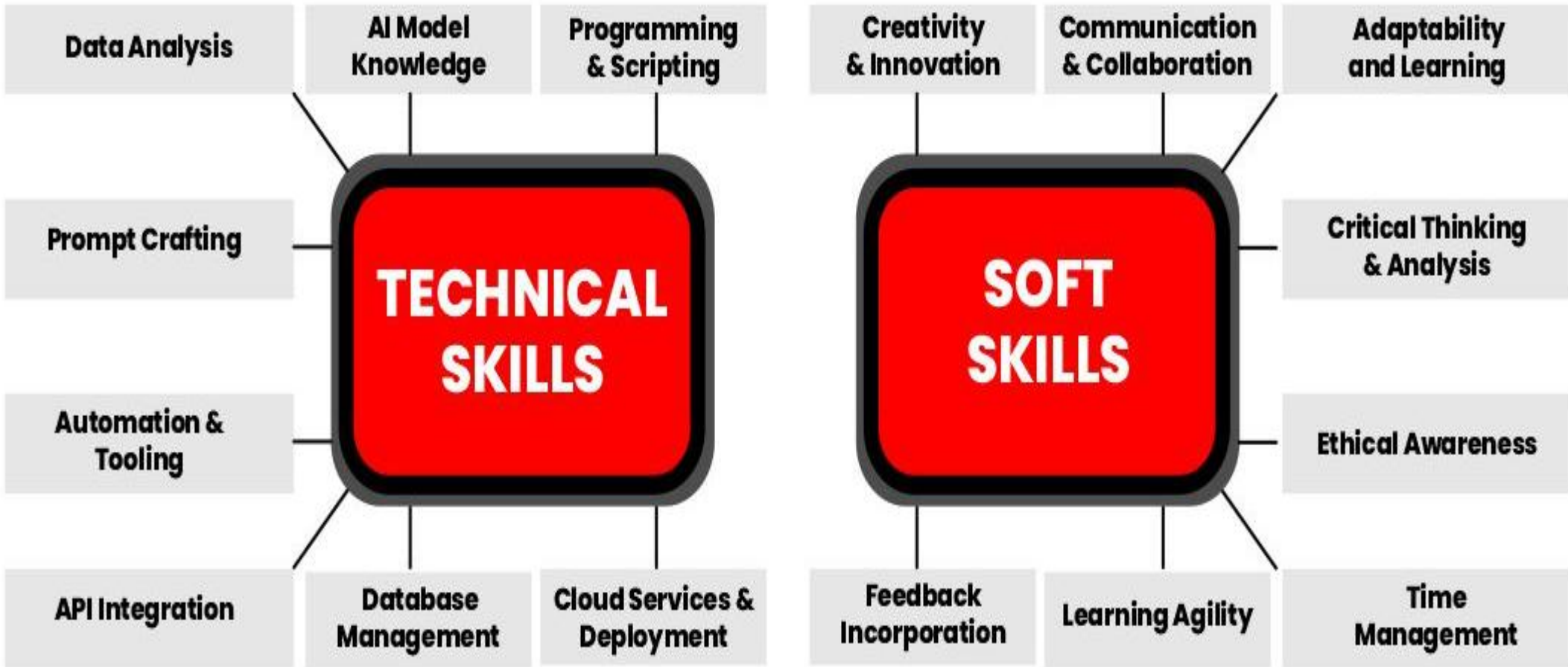
Prompt

Precisely formulated inputs given to AI language model.

Priming

Additional context and instructions to provide superior results.

Key Skills Required for An Effective Prompt Engineer



ME

You

Can you explain the concept of "nostalgia" to a kindergartener?

ChatGPT

Imagine you have a favorite toy or a blanket that you love very much because you've had it since you were a baby. When you hold it or see it, you feel super happy and remember all the fun times you had playing with it or snuggling it when you were even littler. That happy and warm feeling you get thinking about those times is like nostalgia. It's like a big hug from your memories, making you feel good by remembering fun times from before.

FOUNDATIONS

Anatomy of a Prompt

Four components combine to steer a model toward the response you actually want.

1 Instruction

The task you want performed

2 Context

Background info that shapes the answer

3 Input Data

The specific content to act on

4 Output Indicator

Desired format, tone, or structure

example_prompt.txt

```
You are a customer support assistant
for a SaaS billing platform.

A customer says their invoice total
doesn't match their subscription plan
and they were charged twice this month.

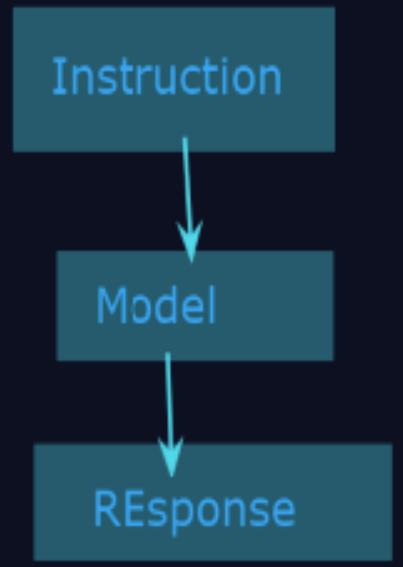
Customer message:
"I was billed $58 instead of $29 —
 please explain and fix this."

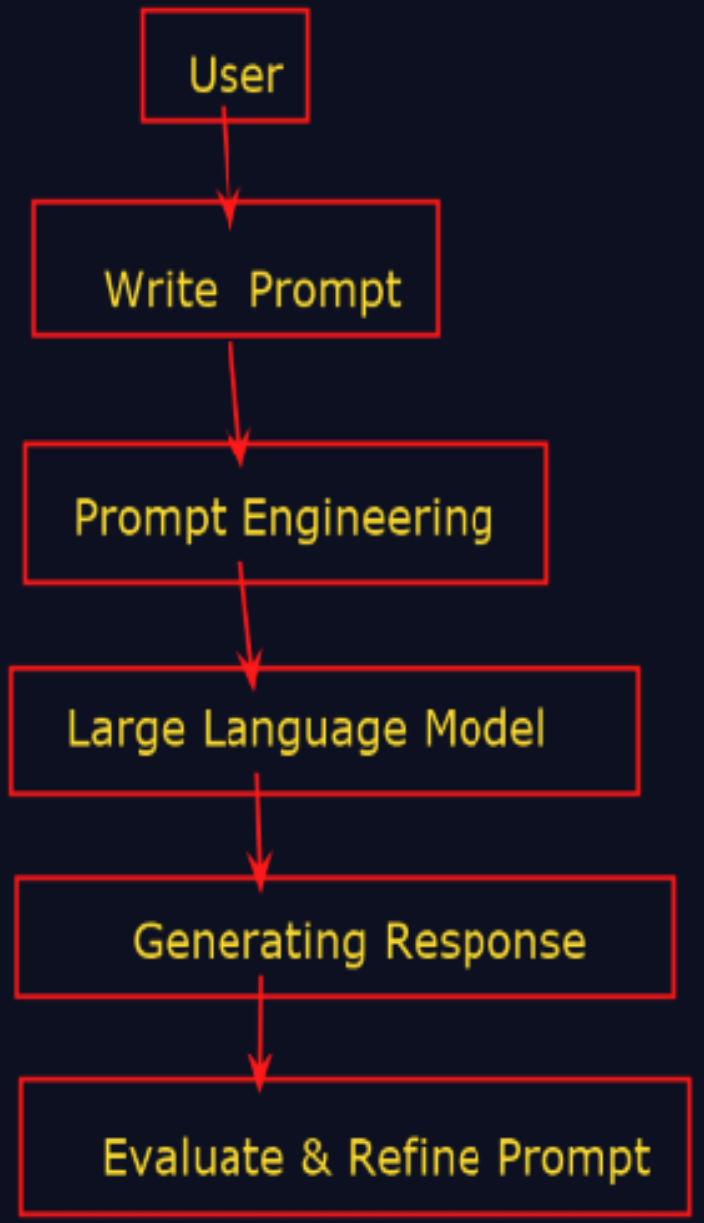
Respond in 3 short sentences, empathetic
tone, and end with a clear next step.
```

to obtain accurate, relavant and high-quality responses.

Zero-shot Prompting : Model solves a task without providing any examples

Application:
-Translation
-summarization
-Question Answering





PROMPTING TECHNIQUES

Zero-Shot Prompting

0

No examples given

The model relies entirely on its pretrained knowledge and the instruction itself to complete a task it has never explicitly been shown examples for.

Best for: simple, well-known tasks — classification, translation, basic summarization.

⚠ Watch out for

Ambiguous instructions, inconsistent formatting, and edge cases the model may misinterpret without a demonstrated pattern.

zero_shot.txt

```
Prompt:
Classify the sentiment of this review
as Positive, Negative, or Neutral.

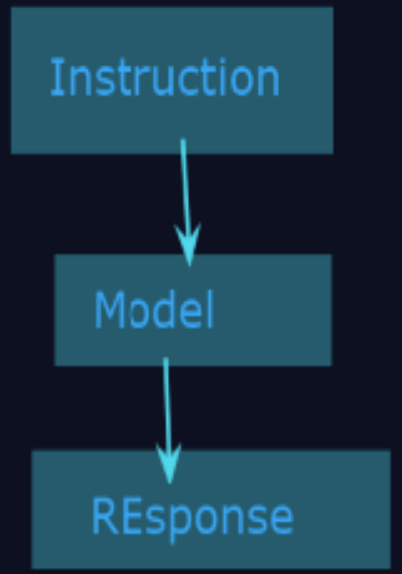
Review: "The battery life is amazing
but the app crashes constantly."

Output:
Sentiment: Mixed / Neutral
Reasoning: Positive hardware feedback
offset by a negative software complaint.
```

to obtain accurate, relavant and high-quality responses.

Zero-shot Prompting : Model solves a task without providing any examples

Application:
-Translation
-summarization
-Question Answering



Chain-of-Thoughts (CoT) Prompting : model is given exactly one example before solving the task

-Application:
-Mathematics
-Logical REasoning
-Programming
-Scientific calculations

One-Short Prompting : atleast one example is required
-Application
-Sentiment analysis

Role Prompting:model is assigned a specific role before answering

-Application
-Eduction
-HEalthcare
-Finance
-Legal

Few-Short Prompting :model receives multiple examples before answering

-Applications
-Classification
-Information Extraction
-Text labelling

RAG :Retrieval - Augmented PRompting

Application
-Enterprise chatbots
-Legal system
-Healthcare
-Research assistant

PROMPTING TECHNIQUES

Few-Shot Prompting

Show the model 2–5 input/output examples so it can infer the pattern, tone, and format you want — before giving it the real task.

few_shot.txt

```
Convert product feedback into a structured tag.

Input: "Shipping took 2 weeks"
Tag: #delivery-delay

Input: "Great value for the price"
Tag: #pricing-positive

Input: "Packaging arrived crushed"
Tag: #packaging-issue

Input: "Customer service resolved my
issue within minutes"
Tag:
#support-positive
```

Consistent formatting

Examples anchor the exact output shape you expect.

Better accuracy

Reduces ambiguity for nuanced or domain-specific tasks.

Trade-off

More tokens per call, and too many examples can bias the model toward the samples shown.

REASONING TECHNIQUES

Chain-of-Thought (CoT) Prompting

Asking the model to reason step-by-step — often triggered with "Let's think step by step" — dramatically improves accuracy on math, logic, and multi-step problems.

✗ Without CoT

Prompt: A store had 120 apples. It sold 35% in the morning and 28 more in the afternoon. How many apples are left?

Answer: **43 apples** *(incorrect — skipped a step)*

✓ With CoT

Prompt: Same question. Let's think step by step.

1. `35% of 120 = 42 sold`
2. `120 - 42 = 78 remain`
3. `78 - 28 = 50 remain`

Answer: **50 apples** *(correct)*

The Role

- Work with cross-functional teams to discuss product development
- Identify uses of AI tools
- Design, develop and refine AI-generated text prompts

Background

- Bachelor's degree in Computer Science or Machine Learning or a related field
- Additional certifications recommended

Skills

- Excellent knowledge of natural language processing
- Knowledge of machine learning
- Comprehensive knowledge of AI-generated content development

Salary

Junior:	$ 280,000
Average:	$ 327,000
Senior:	$ 375,000

FRAMING TECHNIQUES

Role Prompting

Assigning the model a persona, expertise, or point of view focuses its knowledge, vocabulary, and tone toward the audience you need.

S

Security Engineer

"You are a senior application security engineer reviewing code for OWASP Top 10 vulnerabilities."

5

5th-Grade Teacher

"You are a patient elementary school teacher explaining photosynthesis to a 10-year-old."

L

Legal Analyst

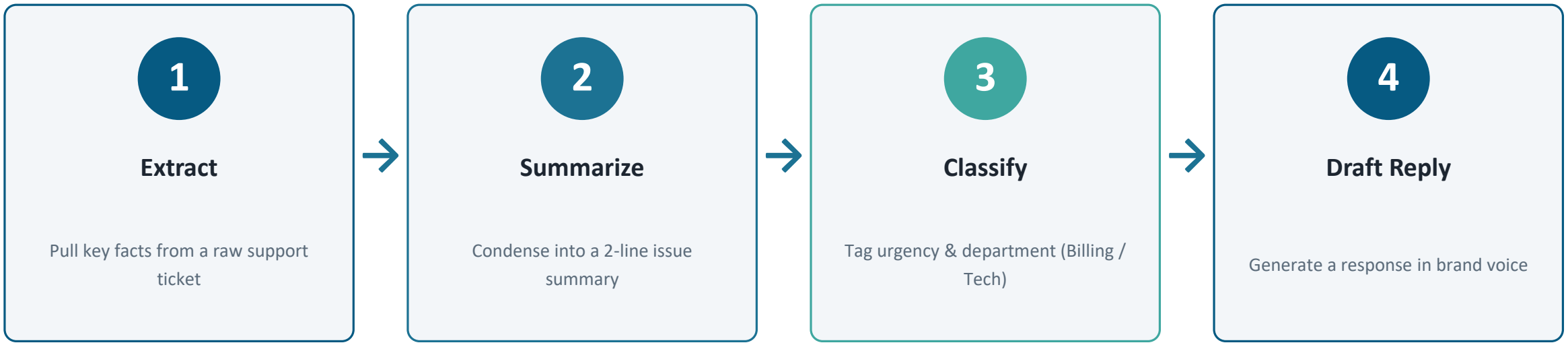
"You are a contracts attorney summarizing risk clauses for a non-legal executive audience."

Why it works: Role framing narrows the model's response space to the vocabulary, priorities, and depth associated with that expertise — improving relevance without changing the underlying task.

COMPOSING PROMPTS

Prompt Chaining

Breaking a complex task into a sequence of smaller prompts — each step's output feeds the next — improves reliability over one giant prompt.



Why chain instead of one prompt? Each link is easier to test, debug, and swap independently; smaller, focused prompts reduce hallucination and make failures easy to localize to a single step.

MACHINE-READABLE OUTPUT

Structured Outputs: JSON & XML

Constraining output to a schema makes LLM responses directly consumable by downstream code — no brittle text parsing required.

prompt.txt

```
Extract the order details below and
return ONLY valid JSON matching:
{"item": str, "qty": int,
 "price_usd": float}

Text: "Ordered 3 wireless mice at
 $14.99 each, ship to NY."

Output:
{
  "item": "wireless mouse",
  "qty": 3,
  "price_usd": 14.99
}
```

{} Reliable parsing

Downstream systems can safely deserialize the response without regex or guesswork.

<> Schema enforcement

Pair with function-calling / JSON-mode APIs or validators (Pydantic, Zod) to guarantee shape.

Tip

State the format explicitly ("return ONLY valid JSON, no prose") and show the exact schema — models mirror formats they are shown.

TESTING & OPTIMIZATION

Tooling: Promptfoo & ChainForge

Purpose-built tools for systematically testing, comparing, and evaluating prompts before they ship to production.

Pf Promptfoo

Open-source CLI/library for prompt testing

- ✦ Define test cases (inputs + expected criteria) in YAML/JSON
- ✦ Run the same prompt across multiple models side-by-side
- ✦ Automated scoring: exact match, LLM-graded rubric, similarity
- ✦ CI/CD integration to catch prompt regressions before deploy

Cf ChainForge

Visual playground for prompt experimentation

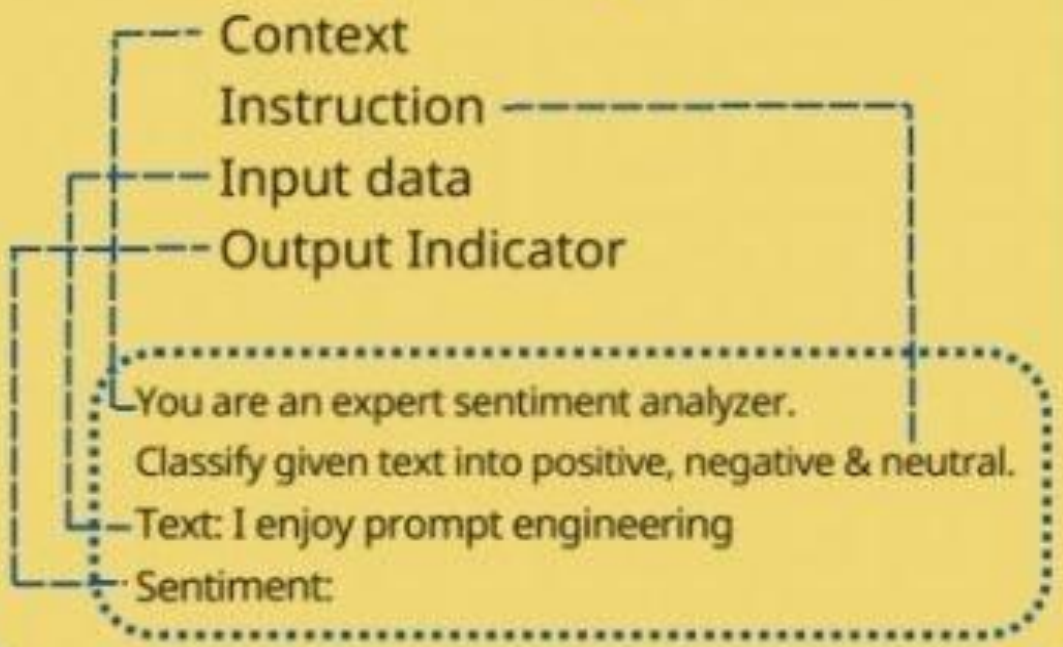
- ✦ Drag-and-drop flow builder — no code required to start
- ✦ Compare prompt variants and models in a visual grid
- ✦ Built-in evaluators + custom scoring scripts (Python/JS)
- ✦ Great for rapid, exploratory iteration before formal testing

Prompt Engineering

(Effective communication & collaboration with AI)

Prompt Engineering is art and science of crafting inputs(prompts) to AI models to get the desired output

Prompt Components



Techniques

Zero-shot
One-shot
Few-shot
Chain of Thought
Self Consistency
Generate Knowledge
Automatic Prompt Engineering
Active Prompt
Directional Stimulus
ReAct
Multimodal CoT
Graph Prompting

Use Cases

Text Summarization
Question Answering
Code Generation
Role Playing
Text Classification
Reasoning
Art Generation
Grammar correction
Bug finding
Language Translation
Idea Generation
& many more

Best Practices

Understand the model's capabilities and limitations
Use clear and specific language
Provide examples and feed

Explain the context in as much detail as possible
Experiment with different formats and styles
Evaluate and refine

PUTTING IT TOGETHER

A Framework for Designing & Optimizing Prompts



Prompt engineering is iterative — treat prompts like code: version them, test them, and measure regressions.



Quick Guide:
AI Prompt Engineering

Strategies for better results in ChatGPT, Perpexity and other AI tools.

PRO TIP
Use adjectives to help define your "role"

Your Prompt should include

WHO:
Role: Who is the tool writing as?
Audience: Who is it writing FOR?

Who is the tool "pretending" to be--a teacher, a scientist, a dancer, a tennis coach? And who is this person writing for? 12 year olds, the board of the Arts Comission, a professional tennis player?

WHAT:
Task: What do you want the tool to do
Voice: Should the tone be informal, chatty, professional?

More detaiils = better results. This takes some planning. Think the ideal response , what you want it to include, how you want it to look , how it should sound, and include that in the instructions.

WHY:
What is the **goal** of the task?

Explaining the **purpose of the task** and what you're **trying to achieve** helps the AI Tool shape its response to better meet your needs.

WHERE:
What is the **context** of the task?

Include necessary **background,** any **limitations** or things NOT to include. Again, this helps generate better results to fit the task.

EXAMPLE

You are an **expert, creative writer, working on an essay about pollution. Create a table** with 3 columns labelled nouns, verbs, adjectitves. **Create a vocabulary list of 20 pollution words** for each column as a word cache for the essay . The essay is **showing the damage to habitat and animals.** It is **factual,** but will create a **sense of crisis.** The vocabulary list should **support those goals.**

To improve your results...

Include examples : show the AI what you want the reponse to look like/include.
Prompt Stacking: Too many directions can confuse the AI. Break the prompt into separate parts. The AI can use results from previous prompts to generate new tasks.
Explain Reasoning: Tell it to explain its thinking step-by-step.
Ask for multiple versions: Lets you pick and choose and build on the better options.

KEY TAKEAWAYS

Design & Optimize Prompts Like You Would Code

- ✓ Start with clear anatomy: instruction, context, input, and output format.
- ✓ Use few-shot examples and CoT reasoning for nuanced or multi-step tasks.
- ✓ Chain prompts and enforce structured output to build reliable pipelines.
- ✓ Evaluate systematically with Promptfoo and ChainForge — don't guess.

Thank You

PROMPT ENGINEERING

Prompt Engineering is the art and science of designing, structuring and refining prompts (instructions) to get accurate, relevant and useful responses from AI models.



THE ANATOMY OF A GOOD PROMPT

ROLE — Assign a role to the AI

+

TASK — Specify what you want

+

CONTEXT — Provide relevant background information

+

CONSTRAINTS — Add limitations (length, style, format, etc.)

+

OUTPUT FORMAT — Define the desired output format

EXAMPLE OF A WELL-ENGINEERED PROMPT

You are an AI tutor.

Task: Explain the concept of Machine Learning.

Context: I am a second-year engineering student with basic programming knowledge.

Constraints: Use simple language. Give examples.

Output Format: Provide the answer in bullet points and include a comparison table.

BEST PRACTICES

- Be clear, specific and unambiguous.
- Provide sufficient context.
- Break complex tasks into steps.
- Specify the desired output format.
- Use examples when needed.
- Iterate and refine the prompt.

TYPES OF PROMPT ENGINEERING (WITH EXAMPLES)

1 ZERO-SHOT PROMPTING

The model performs the task without any examples.

Example:

Translate the following sentence into French.

I love programming.

2 ONE-SHOT PROMPTING

Provide one example before the task.

Example:

Example:
Good → Positive
Bad → Negative

Now classify: Excellent

3 FEW-SHOT PROMPTING

Provide multiple examples to establish a pattern.

Example:

Happy → Positive
Sad → Negative
Amazing → Positive
Terrible → Negative

Now classify: Wonderful

4 CHAIN-OF-THOUGHT PROMPTING

Encourage the model to explain its reasoning step by step.

Example:

Solve this math problem.
Step by step, show your calculations and explain how you arrive at the final answer.

Problem: 23 × 17 = ?

5 ROLE-BASED PROMPTING

Assign a professional role to the AI.

Example:

You are a cybersecurity expert. Explain phishing attacks with real-world examples.

6 INSTRUCTION PROMPTING

Give direct instructions for the task.

Example:

Summarize the following article in five bullet points.

7 CONTEXTUAL PROMPTING

Provide additional context or background information.

Example:

Our company develops healthcare software. Suggest AI applications for patient management.

8 STEP-BY-STEP PROMPTING

Break the task into steps and ask the model to follow them.

Example:

Step 1: Analyze the requirements.
Step 2: Design the algorithm.
Step 3: Write the Python code.
Step 4: Explain the time complexity.

BEFORE vs AFTER

POOR PROMPT	IMPROVED PROMPT
Explain AI.	Explain Artificial Intelligence to Class XII students in 300 words with examples.
Write code.	Write a Python program to implement Merge Sort with comments and analyze its time complexity.
Summarize this.	Summarize the article in five bullet points highlighting the key findings.
Make a presentation.	Create a 10-slide presentation outline on Generative AI for business executives, including speaker notes.

PROMPT ENGINEERING WORKFLOW

Define Objective → Assign Role → Provide Context → Specify Task → Add Constraints → Define Output Format → Review & Refine → Evaluate Output → Iterate if Needed → (back to Define Objective)